# Architecture, system design and infrastructure

*POC on one Mac (Apple M4, 16 GB, macOS) · Open source first · 5 November 2026*

## 1 Architecture at a glance

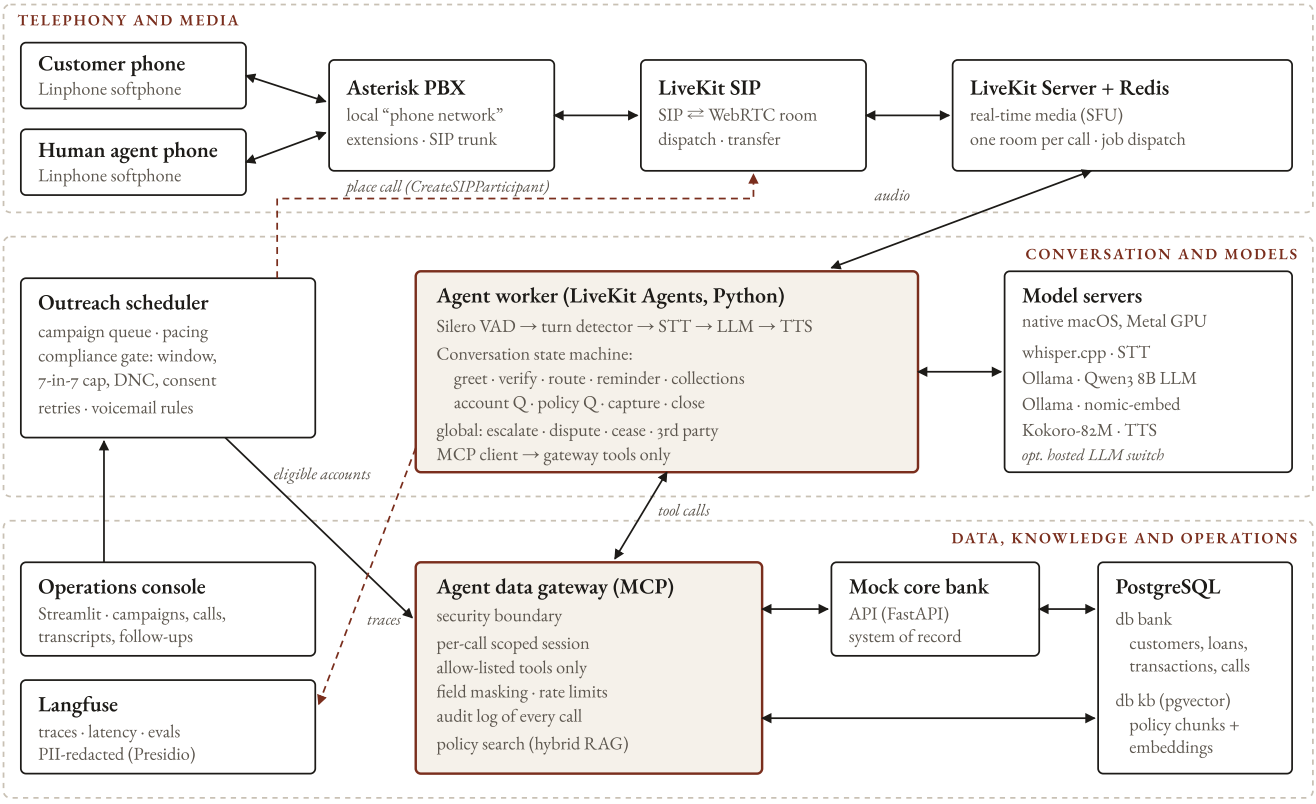



*Figure 1. Components and data flow. Shaded boxes are the two pieces that carry the product logic: the agent worker and the data gateway. Everything else is off-the-shelf infrastructure.*

The design follows the production pattern from Document 1, with three deliberate boundaries. **Media** (LiveKit) is separate from **reasoning** (the agent worker), and reasoning is separate from **data** (the gateway). The agent can reach bank data only through a small set of gateway tools, and only for the customer it has verified on that call. Contact rules are enforced by the scheduler before a call exists, so the model cannot override them.

## 2 Key flows

### *Outbound missed-payment call*

1. The scheduler asks the gateway (service scope) for accounts 1–30 days past due, then filters them through the compliance gate: local calling window, frequency cap, do-not-call list, consent, open disputes and cease-contact flags.
2. For each eligible account it creates a LiveKit room with metadata (`call_id`, `campaign`, `loan_id`) and calls `CreateSIPParticipant`, which routes the call through Asterisk to the customer's softphone.
3. LiveKit dispatches an agent worker into the room. The worker opens a gateway session bound to `call_id`; the customer is still unverified.
4. The agent runs the state machine: AI and recording disclosure, right-party check, then verification through the `verify_identity` tool. If verification succeeds, the gateway binds that one `customer_id` to the session.
5. The agent explains the overdue amount (`get_loan_status`), asks for the reason, explains consequences (`search_policy`), and records the outcome (`record_promise_to_pay` or `request_callback`).

6. For escalations it calls `escalate`, which writes a ticket with a summary, and transfers the call by SIP to the human extension.
7. When the call ends, the worker writes the structured outcome, the transcript is redacted, and traces are flushed to Langfuse.

### *Inbound call*

The customer dials support extension `100`. Asterisk forwards the call to LiveKit SIP, a dispatch rule creates a room, and an agent joins. The caller's number is used only as a *hint* for candidate customers. General policy questions are answered without verification; anything account-specific triggers verification first.

## 3 Agent design

Each state is a LiveKit Agents `Agent` with its own short instructions and only the tools it needs. Hand-offs between states are explicit function returns. Small local models stay on track because each prompt is short and each state has at most three or four tools. Global "interrupt" intents (human, dispute, stop calling, a third party on the line, a lawyer or bankruptcy) are checked on every turn and override the current state.

| GATEWAY TOOL | SCOPE | PURPOSE |
|---|---|---|
| `lookup_by_phone` | unverified | Returns only a masked first name and candidate ids (held by the gateway, never shown to the model). |
| `verify_identity` | unverified | Checks date of birth and the last 4 loan digits. Binds the customer on success; locks after 2 failures. |
| `search_policy` | any | Hybrid retrieval over the knowledge base; returns chunks with `doc_id`, version and section. |
| `get_loan_status` | verified | Product, outstanding balance, next due date and amount, days past due, fees accrued. |
| `get_recent_transactions` | verified | Last 5 payments, with pending payments flagged (for "I already paid"). |
| `get_interaction_history` | verified | Last calls, open promises and open tickets, for continuity. |
| `record_promise_to_pay` | verified | Validates the date against the policy window and the amount against what is due. |
| `request_callback`<br>`escalate` | any | Creates a follow-up task or a human hand-off ticket with reason and summary. |
| `set_contact_flag` | verified | Records a dispute, cease-contact or preferred-time flag. The scheduler honours it immediately. |

> **Gateway security rules.** No free-form queries; every tool has typed inputs. The session token comes from the dispatcher, not the model. Customer ids are never accepted from the model after binding. Responses contain only the fields the state needs, with full account numbers and national IDs masked. Each call is rate-limited and audited (who, what, when, result). Tool outputs are treated as data, not instructions, which guards against prompt injection. The agent can write only to interaction tables, never to balances.

## 4 Mock banking system

A small but realistic stand-in for a core banking system: PostgreSQL holds the data, and a FastAPI service exposes it as the bank's internal API. All data is synthetic, generated with Faker using a fixed random seed so that test scenarios are reproducible.

| TABLE | KEY FIELDS | SEED ROWS |
|---|---|---|
| `customers` | id, full_name, dob, phone, email, address, timezone, segment, kyc_status, preferred_language | 200 |
| `accounts` | id, customer_id, type (current or savings), number, balance, status | 260 |
| `loan_products` | code, name, rate_type, apr_range, term_range, grace_days, late_fee_rule, policy_doc_id | 5 |
| `loans` | id, customer_id, product_code, number, principal, apr, term_months, start_date, status, days_past_due, bucket | 320 |
| `installments` | loan_id, seq, due_date, principal_due, interest_due, fee_due, paid_amount, paid_-date, status | ~9,000 |
| `transactions` | id, account_id, loan_id, ts, amount, channel (branch, mobile, transfer), status (posted or pending), reference | ~12,000 |
| `contact_preferences` | customer_id, consent_voice_ai, dnc, preferred_window, cease_contact, dispute_open | 200 |
| `interactions` | call_id, customer_id, loan_id, direction, campaign, started_at, outcome JSON (see PRD §8), transcript_uri, versions | grows |
| `promises_to_pay`, `tasks` | Promises (date, amount, kept or broken); callbacks and escalations (type, due, owner, status) | grows |
| `audit_log` | ts, call_id, tool, args_hash, customer_id, decision, latency_ms | grows |

**Loan products:** personal loan, auto loan, home mortgage, small-business working capital, and a salary-advance microloan. Each has different grace periods and fee rules, so policy answers must be product-specific.

### *Scripted test customers*

Alongside the random population, about 12 hand-crafted personas each exercise one path. Their phone numbers map to softphone accounts, so they can be dialled live.

| PERSONA | SITUATION | EXPECTED OUTCOME |
|---|---|---|
| **On-time Olivia** | Due in 2 days, never late | Reminder, intent to pay confirmed |
| **Forgetful Frank** | 5 days past due, first miss | Reason “forgot”, promise in 3 days |
| **Hardship Hana** | 22 days past due, lost job | Hardship referral, escalate to human |
| **Disputing Dawit** | Claims the fee is wrong | Dispute flag set, collection paused, human |
| **Paid-already Priya** | Payment pending, not yet posted | Payment in transit, no promise needed |
| **Spouse answers** | Third party picks up | Neutral message, no details given |
| **Stop-calling Sam** | Asks never to be called | Cease flag; scheduler excludes him |
| **Curious inbound caller** | Asks about early repayment | Policy answer with citation |

**Mock core bank API** (internal, called only by the gateway and scheduler): `GET /customers/{id}`, `GET /customers?phone=`, `GET /loans/{id}`, `GET /loans/{id}/installments`, `GET /accounts/{id}/transactions`, `GET /collections/eligible?bucket=1-30`, `POST /interactions`, `POST /promises`, `POST /tasks`, `PATCH /contact-preferences/{id}`.

## 5 Policy knowledge base (embedded)

About 20 synthetic Markdown policy documents, each with front-matter (`doc_id`, `title`, `product`, `version`, `effective_date`, `audience`):

- Product terms, one per loan product
- Repayment schedules and due-date rules
- Grace periods, late fees and penalty interest
- Delinquency stages and consequences (reminders, fee, bureau reporting at 30 days past due, collections at 60, default at 90)
- Hardship assistance and restructuring (eligibility; humans decide)
- Promise-to-pay rules (maximum 14 days, partial payments)
- Payment channels and posting times
- Early repayment and payoff
- Disputes and error resolution
- Identity verification and privacy procedure
- Complaints and escalation procedure
- Contact and calling rules (also read by the scheduler as configuration)
- Required disclosures (versioned script text)
- Frequently asked questions

**Ingestion:** documents are split by heading into chunks of about 300–500 tokens with section paths, embedded with `nomic-embed-text` (768-dimensional) via Ollama, and stored in PostgreSQL with `pgvector` (HNSW index) alongside a `tsvector` column for keyword search. **Retrieval:** hybrid vector and full-text search merged with reciprocal rank fusion, filtered by product and effective date, returning the top 4 chunks. If the best score is below a threshold, the agent says it will follow up rather than guessing. Re-ingesting a new document version keeps the old version for audit. Here "embedded" means vectorised; if you meant an in-process store with no server, LanceDB or `sqlite-vec` fit the same interface.

## 6 Infrastructure list: what we need and why

| COMPONENT | CHOICE · LICENCE | RUNS AS | WHY WE NEED IT |
|---|---|---|---|
| **Container runtime** | Colima (MIT) or Docker Desktop | native | Runs the Linux services (Postgres, Redis, Asterisk, LiveKit SIP, Langfuse) reproducibly with one `docker compose up`. |
| **Local PBX** | Asterisk (GPLv2) | Docker | Stands in for the phone network: softphone extensions for customers and human agents, and a SIP trunk to LiveKit. It also lets us test transfers. A carrier trunk can replace it later. |
| **Softphones** | Linphone (GPLv3) | Mac / phone | Real call audio on both ends, playing the customer and the human officer. Free, with no phone numbers or carrier costs. |
| **Media server** | LiveKit Server (Apache 2.0) | native (brew) | Low-latency real-time audio, one room per call, and dispatch of agent workers to calls. |
| **SIP bridge** | LiveKit SIP (Apache 2.0) | Docker | Converts SIP calls into LiveKit rooms: inbound dispatch rules, outbound dialling and call transfer. |
| **Message bus** | Redis / Valkey (BSD) | Docker | Required by LiveKit SIP to coordinate with the LiveKit server. |
| **Agent framework** | LiveKit Agents, Python (Apache 2.0); Silero VAD (MIT); LiveKit turn detector (open weights) | native | Production-grade voice pipeline: streaming, barge-in, turn detection, multi-agent hand-offs, MCP tools, metrics, and text-mode testing. Pipecat (BSD-2) is the fallback. |
| **Speech-to-text** | whisper.cpp server (MIT), large-v3-turbo or small.en | native, Metal | Accurate English transcription on the GPU. It runs natively because Docker on a Mac cannot use the GPU. |

| COMPONENT | CHOICE · LICENCE | RUNS AS | WHY WE NEED IT |
|---|---|---|---|
| **LLM and embeddings** | Ollama (MIT) with Qwen3 8B (Apache 2.0) and nomic-embed-text (Apache 2.0) | native, Metal | Local reasoning and tool calling with an OpenAI-compatible API; also produces the knowledge-base embeddings. No data leaves the machine. |
| **Text-to-speech** | Kokoro-82M via Kokoro-FastAPI (Apache 2.0) | native | Natural, fast, streaming English voices with an OpenAI-compatible endpoint. Piper is archived, so it is not recommended. |
| **Database and vector store** | PostgreSQL 16 + pgvector (PostgreSQL licence) | Docker | One engine for the mock core bank, the call outcomes and the embedded knowledge base, with transactional and SQL-auditable storage. |
| **Mock core bank API** | FastAPI (MIT), SQLAlchemy, Faker | native | Simulates the bank's system of record and its internal API, so the agent integrates as it would with a real bank. |
| **Agent data gateway** | MCP Python SDK (MIT) on FastAPI | native | The secured, controlled gate: scoped sessions, allow-listed tools, masking, rate limits and audit. MCP keeps it reusable by other agents. |
| **PII redaction** | Microsoft Presidio (MIT) | library | Strips names, numbers and dates of birth from transcripts and traces before they are stored or viewed. |
| **Scheduler / dialer** | Python + APScheduler (MIT) | native | Runs campaigns, applies the compliance gate, paces calls, and handles retries and callbacks. |
| **Operations console** | Streamlit (Apache 2.0) | native | A quick internal UI to trigger calls, watch outcomes, read transcripts and work the follow-up queue. |
| **Observability** | Langfuse, self-hosted (MIT) | Docker | Per-turn traces of latency, tool calls and retrieved sources, plus evaluation scores. It has a native LiveKit integration. Arize Phoenix is a lighter alternative. |
| **Test harness** | pytest + LiveKit Agents test utilities | native | Simulated-caller scenarios in text mode with an LLM judge, giving repeatable regression tests for the guardrails. |
| **Optional: hosted LLM** | e.g. Claude Haiku 5.5 (proprietary API) | cloud | Benchmark for quality and latency against the local model. Synthetic data only. |
| **Optional: carrier trunk** | Twilio / Telnyx Elastic SIP | cloud | Real phone numbers and calls to mobile phones, when the POC moves beyond the laptop. |

### *Resource budget on the Mac (approximate)*

| COMPONENT | MEMORY | NOTE |
|---|---|---|
| **Qwen3 8B (Q4_K_M) in Ollama** | ~5.5 GB | Disable "thinking" mode for latency; Qwen3 4B is the fallback. |
| **whisper.cpp (large-v3-turbo)** | ~1.6 GB | small.en (~0.5 GB) if latency is tight. |
| **Kokoro TTS + embeddings model** | ~0.8 GB | |
| **Docker VM: Postgres, Redis, Asterisk, SIP** | ~1.5 GB | |
| **Langfuse stack (optional)** | ~2.5 GB | Start only when tracing is needed, or use Phoenix. |
| **Agent, gateway, API, console (Python)** | ~1 GB | Total is about 13 GB with everything running, which fits in 16 GB. |

**Known risks.** SIP and RTP over Docker networking on macOS can be fiddly. We publish a fixed RTP port range and pin the advertised IP, and can fall back to running Asterisk natively with Homebrew. LiveKit's browser playground is a no-telephony fallback for testing the agent from day one.

## 7 From POC to production

| | |
|---|---|
| **Telephony** | Swap Asterisk for a carrier SIP trunk, register numbers, add answering-machine detection and caller-ID reputation (STIR/SHAKEN). |
| **Scale** | LiveKit cluster or LiveKit Cloud; agent workers autoscaled on Kubernetes; GPU inference servers (vLLM) or managed STT, LLM and TTS to reach about 800 ms. |
| **Bank integration** | Point the gateway at the real core-banking and CRM APIs behind the bank's API gateway with mTLS and OAuth service identities. The agent's code is unchanged. |
| **Governance** | Jurisdiction-specific rules in the policy configuration, legal review of disclosures, retention policies, encrypted recordings, sampled human QA, and a model-risk review. |